

Retail Data Pipeline Workflow

End-to-End Analytics with Microsoft Azure & Fabric

Step 1: Data Source Setup in Microsoft Azure

The workflow begins in the Azure cloud environment. Raw operational data from various retail systems is securely deposited into an Azure Blob Storage container named **source**. This acts as our primary data lake drop-zone, housing heterogeneous file formats including JSON, CSV, and Excel spreadsheets.

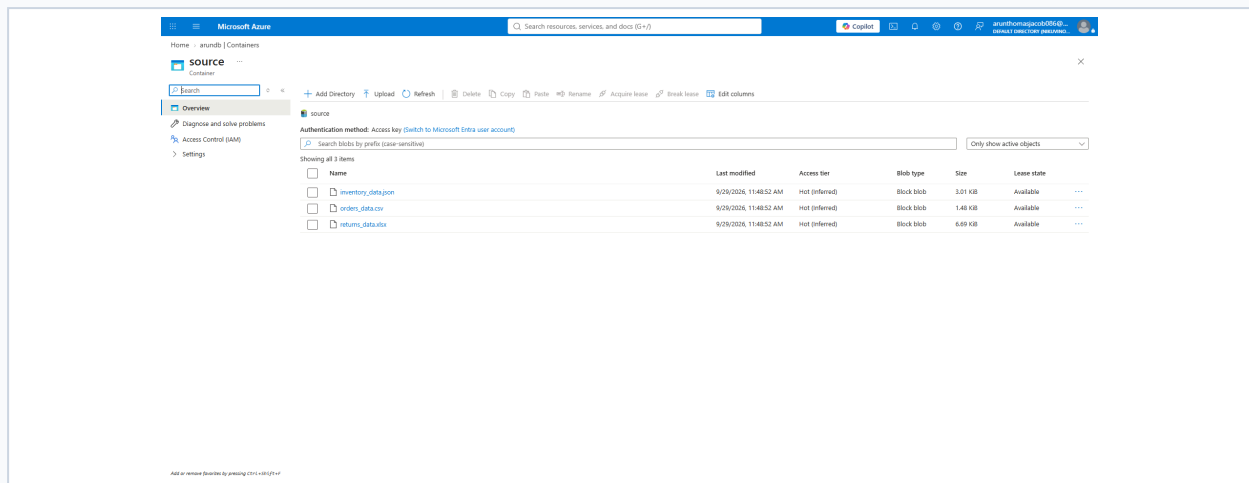
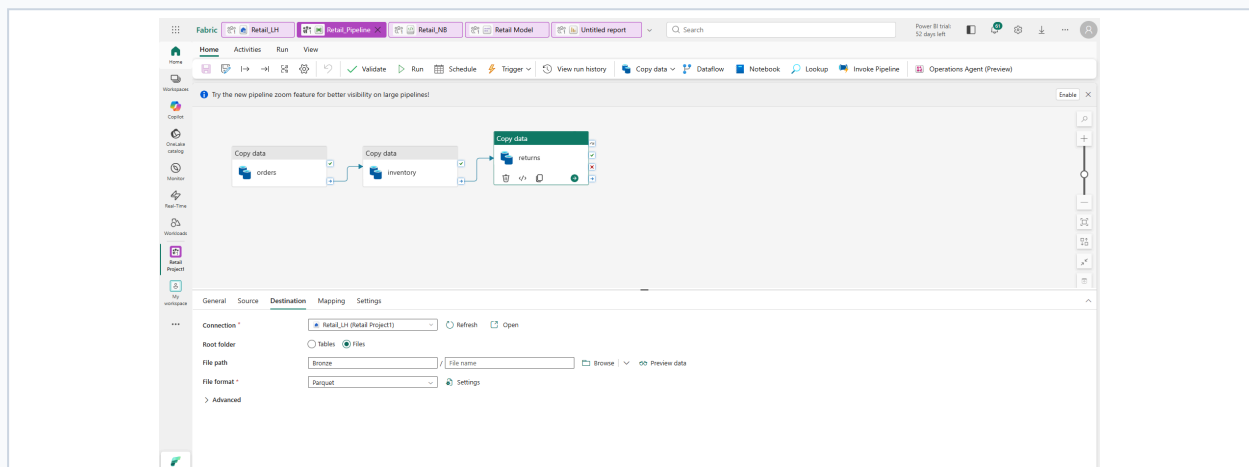


Fig 1.1 - Azure portal showing inventory, orders, and returns data files.

Step 2: Data Ingestion via Microsoft Fabric Pipelines

Microsoft Fabric Pipelines orchestrate the automated movement of data from Azure Storage into the analytical environment. The **Retail_Pipeline** executes parallel Copy Data activities, securely extracting data and landing it in the Fabric Lakehouse (Retail_LH). The destination is set to the Bronze tier, automatically converting files into optimized Parquet format for downstream processing.



Step 3: Medallion Architecture Storage

The Retail_LH lakehouse implements a standard Medallion architecture. Raw ingested Parquet files reside in the **Bronze** folder. As data is cleaned, validated, and enriched, it is promoted to the **Silver** and ultimately **Gold** folders, ensuring high data quality for business reporting.

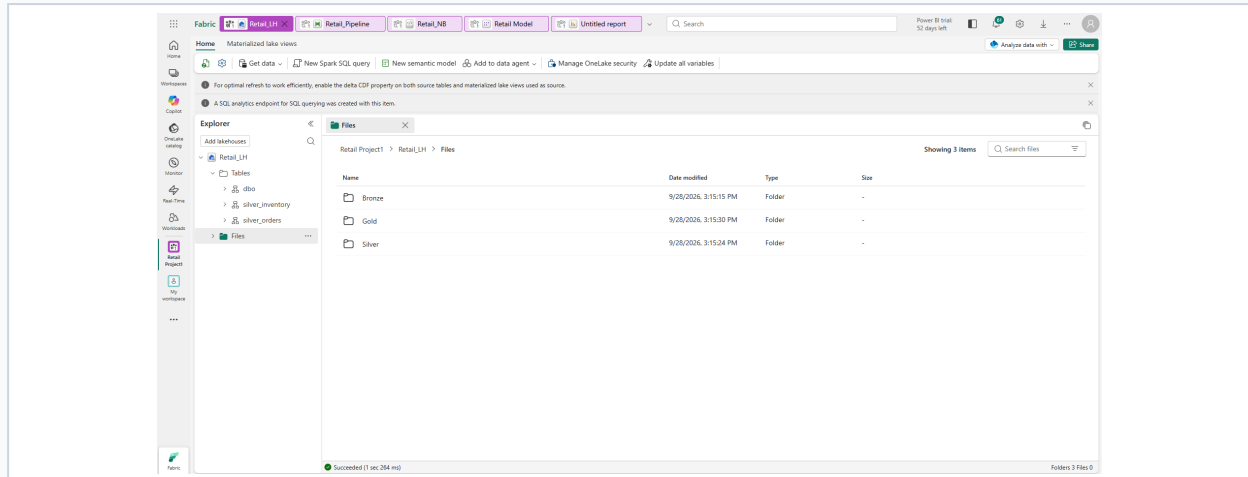


Fig 3.1 - OneLake file explorer showcasing the Bronze, Silver, and Gold organizational structure.

Step 4: Data Transformation with PySpark Notebooks

Data cleansing and transformation are executed using Fabric Notebooks (Retail_NB). Utilizing PySpark, data engineers load the raw Parquet data from the OneLake path, perform schema enforcement, handle null values, and standardize columns.

The output below demonstrates the normalized inventory view, making critical fields like product_id, cost_price, and warehouse availability ready for analytical modeling.

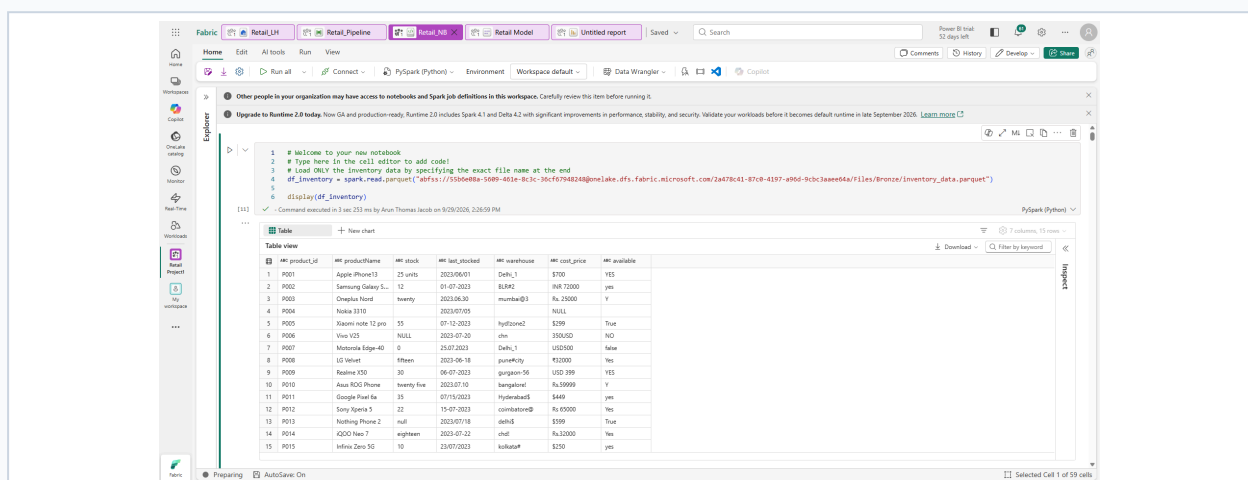


Fig 4.1 - Interactive notebook execution displaying the cleaned tabular inventory data.

Step 5: Semantic Modeling

Once data reaches the Gold layer, a Semantic Model (Retail_Model) is constructed. This model defines relationships between tables and centralizes business logic. The gold_retail_joined table acts as the core view, surfacing pre-calculated measures such as *Estimated_Gross_Margin* and *Cost_Price* to ensure all downstream reports share a single source of truth.

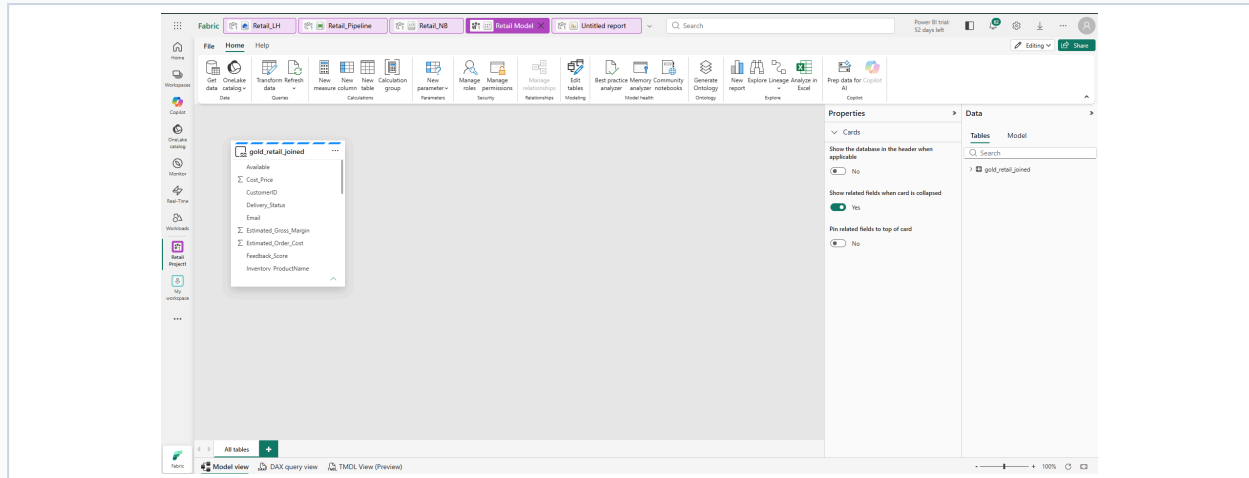


Fig 5.1 - Semantic model view highlighting the fields available for visualization.

Step 6: Executive Dashboard & Reporting

The final consumption layer is an interactive Power BI dashboard designed for retail executives. It connects directly to the Semantic Model to provide near-real-time visibility into operations.

Key Performance Indicators

Total Revenue	167.17K
Total Orders	13
Gross Margin	-163.12K
Inventory Value	4.84M

The dashboard tracks monthly revenue trends, breaks down revenue and stock by individual product lines, monitors delivery success rates, and categorizes return reasons to identify supply chain friction points.

